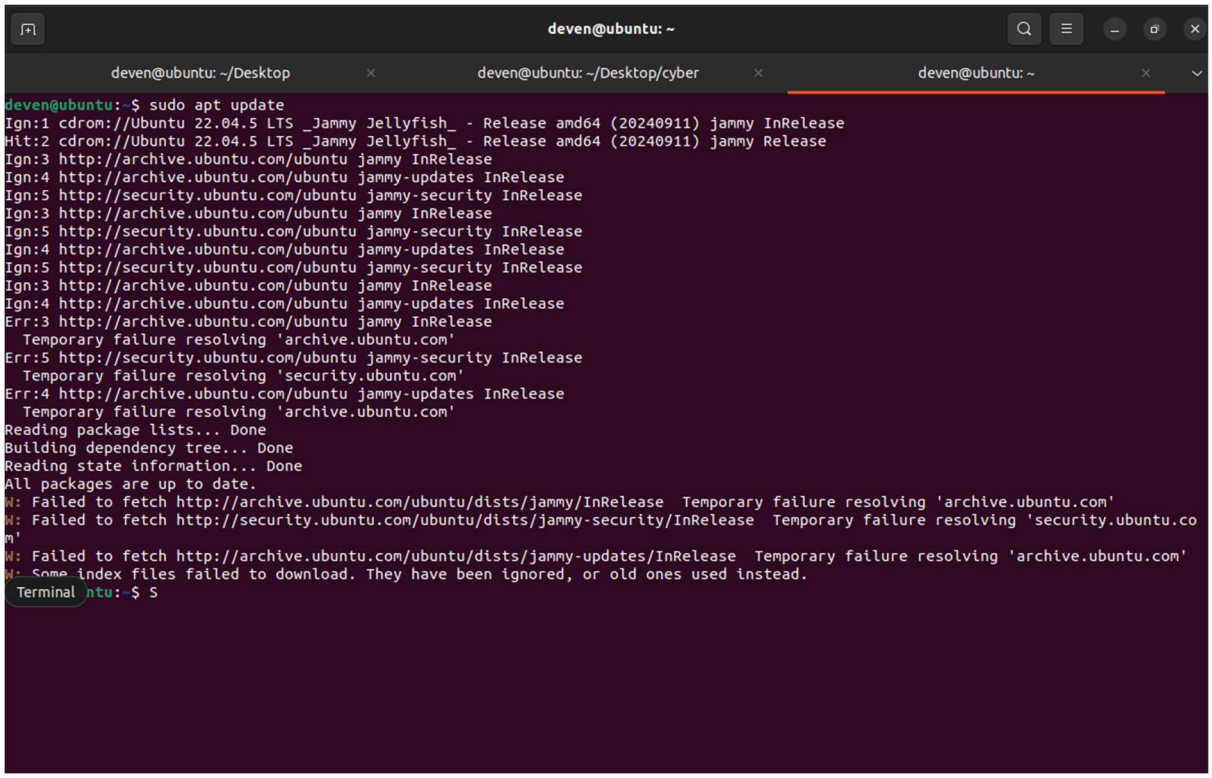
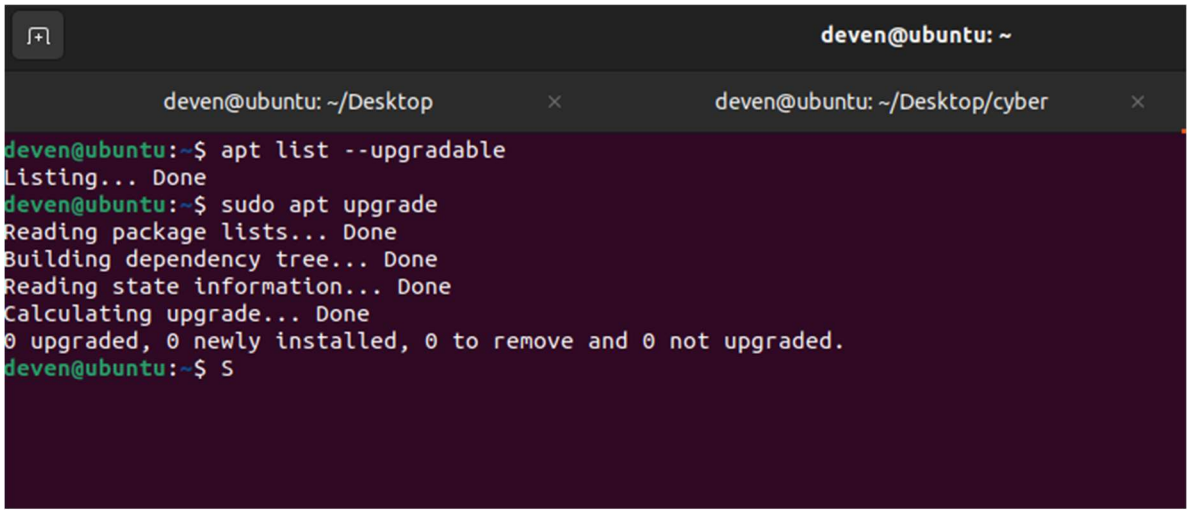
Linux Security Basics

General System Tasks:

1. To retrieve available updates for the system you would first retrieve the information about the available packages by entering the command “sudo apt update”:



2. You can then list the packages with available upgrades with “”apt list –-upgradable” and upgrade them with “sudo apt upgrade”



3. After updating to reboot the system simply input “sudo reboot”.

User Tasks:

4. Changing the current user to root using “sudo su root”. When logging in as root the terminal asks for a password. This gives the user superuser privleges, which give the user unrestricted access to all commands, files, directories, resources and conrtol of permissions.

```
deven@ubuntu: ~/Desktop
deven@ubuntu:~$ sudo su root
[sudo] password for deven:
root@ubuntu:/home/deven# SSS
```

5. The useradd command does **NOT** create a directory and password for a user. The adduser command prompts you through setting up the directory and password. Here’s an example of adding bobby using useradd:

```
root@ubuntu: /home/de
deven@ubuntu: ~/D...    deven@ubuntu: ~/D...
root@ubuntu:/home/deven# useradd Bobby
root@ubuntu:/home/deven#
```

The following is an example of adduser with sally.

```
deven@ubuntu: ~/Desktop    deven@ubuntu: ~/
root@ubuntu:/home/deven# adduser sally
Adding user `sally' ...
Adding new group `sally' (1006) ...
Adding new user `sally' (1005) with group `sally' ...
Creating home directory `/home/sally' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
Changing the user information for sally
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
root@ubuntu:/home/deven#
```

6. To change the current user you use the command “su [username]” This changes the from the root@ubuntu to sally@ubuntu.

```
deven@ubuntu: ~/Desktop
root@ubuntu:/home/deven# su sally
sally@ubuntu:/home/deven$
```

7. Since sally is does not have supersuer privlidges when trying to create a user sally is denied access. To give sally superuser abilities, we can simply add her to the sudo group with “sudo usermod -a -G sudo sally”

```
deven@ubuntu: ~/Desktop
root@ubuntu:/home/deven# su sally
sally@ubuntu:/home/deven$ useradd earl
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
```

8.After logging back into my account we want to delete Bobby. To delete a user you use the command “sudo userdel [username]”

```
deven@ubuntu: ~/Desktop
sally@ubuntu:/home/deven$ exit
exit
root@ubuntu:/home/deven# exit
exit
deven@ubuntu:~$ sudo userdel Bobby
deven@ubuntu:~$
```

9.To change sally’s password we use “sudo passwd sally”. Due to creating an easy to remember password and this simply being an exercise. I got bad password warnings for not inputting an appropriate password for good security.

```
deven@ubuntu:~$ sudo passwd sally
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
deven@ubuntu:~$
```

10.Even though it is easier to complete tasks/commands when logged in as root. It is bad practice because while logged in as root the specific person making changes to the system is not captured, where when a user is logged in as themselves, there’s a record of what was changed and by who. Not only this but it becomes easier to make catastrophic changes to your system because there’s

not safeguards before taking an action, such as accidentally installing malicious software, deleting a directory or modifying crucial files.

11. To see what your user id is simply enter the command “id”. Uid= your id number.

```
deven@ubuntu:~$ id
uid=1003(deven) gid=1004(deven) gro
e)
deven@ubuntu:~$
```

Group Tasks:

12. To see which groups a user belongs to we can simply use “groups [username]”, the following is which groups Ubuntu belongs to.

```
deven@ubuntu:~$ groups ubuntu
ubuntu : ubuntu adm cdrom sudo dip plugdev lpadmin lxd sambashare
deven@ubuntu:~$
```

13. Now we can add sally to the sudo group to give her sudo commands like how we stated before “sudo usermod -a -G sudo sally”. We’ll create a new user while logged in as sally. Can’t forget to use sudo useradd!

```
deven@ubuntu:~$ groups sally
sally : sally
deven@ubuntu:~$ sudo usermod -a -G sudo sally
deven@ubuntu:~$ groups sally
sally : sally sudo
deven@ubuntu:~$ su sally
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sally@ubuntu:/home/deven$ useradd brian
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
sally@ubuntu:/home/deven$ sudo useradd brian
[sudo] password for sally:
```

14. We can create a group called cybersec by using the command “sudo groupadd cybersec”.

```
sally@ubuntu:/home/deven$ sudo groupadd cybersec
[sudo] password for sally:
sally@ubuntu:/home/deven$
```

15. Now we can add sally to cybersec similarly to how we added sally to the sudo group.

16. We can check we did it correctly by using from a pool of commands that show groups users are in. We can input “groups sally” which will list the groups sallys in. We can try “id sally” which will show the group ids along with the groups she belongs to. We can also try “grep sally /etc/group” which shows all groups and their members that sally is a part of.

```
deven@ubuntu:~$ sudo usermod -a -G cybersec sally
[sudo] password for deven:
deven@ubuntu:~$ groups sally
sally : sally sudo cybersec
deven@ubuntu:~$ id sally
uid=1005(sally) gid=1006(sally) groups=1006(sally),27(sudo),1008(cybersec)
deven@ubuntu:~$ grep sally /etc/group
sudo:x:27:ubuntu,deven,sally
sally:x:1006:
cybersec:x:1008:sally
```

Permission and Access Contril Lists:

17. To make a new directory we use the “mkdir” command. To check the permissions of a directory we use “ls -ld d[irectory name]”. We’ll make a directory called lab1.

```
deven@ubuntu:~$ mkdir lab1
deven@ubuntu:~$ ls -ld lab1
drwxrwxr-x 2 deven deven 40 Feb 19 23:50 lab1
deven@ubuntu:~$
```

Based on the screenshot you can tell deven is the owner and group owner of the directory. The owner and group can read write and execute, where other can only read and execute.

18. To change directories we use “cd” and to create and edit a bash file we use “nano”. I named the file helloworld.sh. We must include *!/bin/bash at the beginning of the file. The code reads

#!/bin/bash

Echo -n "Hello, world!"

To make the file executable we input “chmod u+x helloworld.sh”. To execute we need only input “./helloworld.sh”

```
deven@ubuntu:~$ cd lab1
deven@ubuntu:~/lab1$ nano helloworld.sh
deven@ubuntu:~/lab1$ chmod u+x helloworld.sh
deven@ubuntu:~/lab1$ ./helloworld.sh
Hello World!deven@ubuntu:~/lab1$
```

19. We enter ”ls -l helloworld.sh” to see permissions for the file. To give the group executing and writing permissions we input “chmod g+xw helloworld.sh”

```
deven@ubuntu:~/lab1$ ls -l helloworld.sh
-rwxrw-r-- 1 deven deven 36 Feb 20 00:11 helloworld.sh
deven@ubuntu:~/lab1$ chmod g+xw helloworld.sh
deven@ubuntu:~/lab1$ ls -l helloworld.sh
-rwxrwxr-- 1 deven deven 36 Feb 20 00:11 helloworld.sh
deven@ubuntu:~/lab1$
```

20. To view the ACL(Access Control List) of the file we use “getfacl”

```
deven@ubuntu:~/lab1$ getfacl helloworld.sh
# file: helloworld.sh
# owner: deven
# group: deven
user::rwx
group::rwx
other::r--

deven@ubuntu:~/lab1$
```

21. To give sally the ability to read and write to the file we use “setfacl -m u:sally:rw- helloworld.sh”. The m marks to modify. The u means user which is why sally follows. rw- gives reading and writing privileges only not executing. Lastly is the file name.

```
deven@ubuntu:~/lab1$ setfacl -m u:sally:rw- helloworld.sh
deven@ubuntu:~/lab1$ getfacl helloworld.sh
# file: helloworld.sh
# owner: deven
# group: deven
user::rwx
user:sally:rw-
group::rwx
mask::rwx
other::r--

deven@ubuntu:~/lab1$
```